

Divisibility Rules

Number World Study Notes

Quick tests for 2, 3, 4, 5, 6, 7, 8, 9, and 10 — with the reasoning behind them

Divisibility rules are shortcuts that let you check whether one number divides another exactly, without doing long division. They are extremely useful for simplifying fractions, factoring numbers, and solving problems quickly. This chapter explains the rule for each number from 2 to 10 and why each rule works.

Key Definitions

Divisibility

A number 'a' is divisible by a number 'b' if dividing a by b gives a remainder of 0, i.e., b is a factor of a.

Why Rules Work (Place Value)

Rules for 2, 4, 5, 8, 10 depend only on the last few digits because 10, 100, and 1000 are themselves divisible by 2, 4, 8, 10 respectively. Rules for 3 and 9 use digit sums because 10 leaves remainder 1 when divided by 3 or 9.

The Nine Divisibility Rules

- By 2: the last digit is 0, 2, 4, 6, or 8 (i.e., even). Reason: 10 is divisible by 2, so only the ones digit matters.
- By 3: the sum of all digits is divisible by 3. Reason: $10 = 9 + 1$, and 9 is divisible by 3, leaving only digit sums to check.
- By 4: the number formed by the last two digits is divisible by 4. Reason: 100 is divisible by 4.
- By 5: the last digit is 0 or 5. Reason: 10 is divisible by 5, so only the ones digit matters.
- By 6: the number is divisible by BOTH 2 and 3 (since $6 = 2 \times 3$).
- By 7: double the last digit, subtract it from the rest of the number; repeat until you can judge divisibility by 7.
- By 8: the number formed by the last three digits is divisible by 8. Reason: 1000 is divisible by 8.
- By 9: the sum of all digits is divisible by 9. Reason: 10 leaves remainder 1 when divided by 9.
- By 10: the last digit is 0. Reason: 10 itself is divisible by 10.

Worked Examples

Example 1: Is 4536 divisible by 3?

Add the digits: $4 + 5 + 3 + 6 = 18$.

18 is divisible by 3 ($18 \div 3 = 6$).

Answer: Yes, divisible by 3

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Example 2: Is 7128 divisible by 8?

Look at the last three digits: 128.

$128 \div 8 = 16$ exactly.

Answer: Yes, divisible by 8

Example 3: Is 2345 divisible by 7?

Double the last digit (5): 10. Subtract from the rest (234): $234 - 10 = 224$.

Repeat: double last digit (4) = 8. Subtract from rest (22): $22 - 8 = 14$.

14 is divisible by 7.

Answer: Yes, divisible by 7

Example 4: Is 6300 divisible by 6?

Check divisibility by 2: last digit 0 is even. Yes.

Check divisibility by 3: digit sum $6+3+0+0 = 9$, divisible by 3. Yes.

Divisible by both 2 and 3, so divisible by 6.

Answer: Yes, divisible by 6

Practice Questions

- 1 Is 5124 divisible by 4?
- 2 Is 8934 divisible by 9?
- 3 Is 1001 divisible by 7?
- 4 Which numbers from 2 to 10 divide 360 exactly?
- 5 Is 47632 divisible by 8?

Answer Key

- 1 Yes. Last two digits 24, and $24 \div 4 = 6$.
- 2 No. Digit sum $= 8+9+3+4 = 24$, which is not divisible by 9.
- 3 Yes. $100 - (2 \times 1) = 98 = 7 \times 14$, divisible by 7.
- 4 2, 3, 4, 5, 6, 8, 9, 10 (not 7).
- 5 Yes. Last three digits 632, and $632 \div 8 = 79$.